	Marwadi University **Faculty of Engineering & Technology** **Department of Information and Communication Technology**	
Subject: Digital Signal and Image Processing (01CT1513)	**Aim:** Simulate discrete time sequences.	
Experiment No: 1	**Date: 19-07-2026**	**Enrollment No:92400133156**

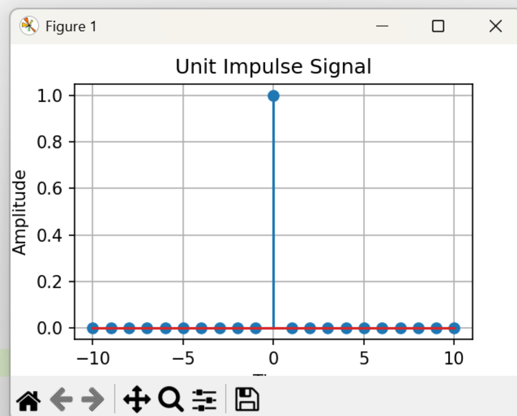
Aim: Simulate discrete time sequences.

Github repo : https://github.com/Diyakalaria19/DSIP_lab_exps

1. Unit Impulse signal

experiment_1 > 1.py > ...

```python
import numpy as np
import matplotlib.pyplot as plt
def unit_impulse(length, position): #here we are defining function
    signal = np.zeros(length)
    signal[position] = 1
    return signal
start = -10
stop = 10
step = 1
x = np.arange(start, stop+step, step)
impulse_signal = unit_impulse(len(x), abs(start)//step)
plt.stem(x, impulse_signal)
plt.xlabel('Time')
plt.ylabel('Amplitude')
plt.title('Unit Impulse Signal')
plt.grid(True)
plt.show()
```



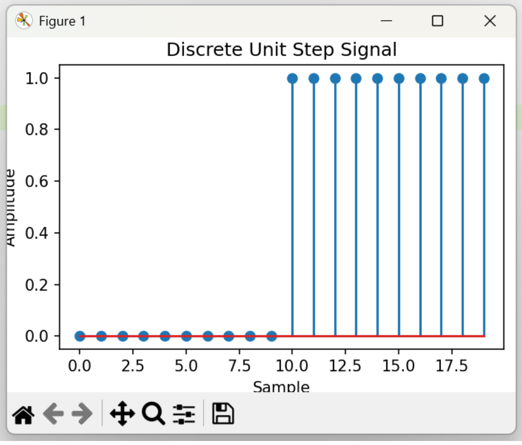
2. Unit Step signal

experiment_1 > 2.py > ...

```python
import numpy as np
import matplotlib.pyplot as plt

def simulate_discrete_unit_step(num_samples):
    unit_step = np.zeros(num_samples)
    unit_step[num_samples // 2:] = 1
    return unit_step

num_samples = 20
discrete_unit_step = simulate_discrete_unit_step(num_samples)
plt.stem(discrete_unit_step)
plt.title('Discrete Unit Step Signal')
plt.xlabel('Sample')
plt.ylabel('Amplitude')
plt.tight_layout()
plt.show()
```



3. Ramp signal

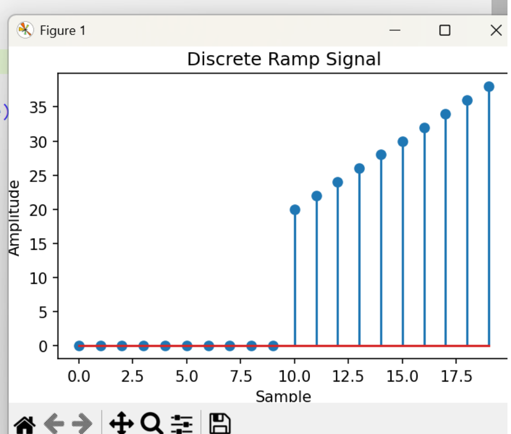
experiment_1 > 3.py > ...

```python
import numpy as np
import matplotlib.pyplot as plt

def simulate_discrete_ramp(num_samples, slope):
    ramp = np.zeros(num_samples)
    ramp[num_samples // 2:] = slope * np.arange(num_samples // 2, num_samples)
    return ramp

num_samples = 20
slope = 2
discrete_ramp = simulate_discrete_ramp(num_samples, slope)

plt.stem(discrete_ramp)
plt.title('Discrete Ramp Signal')
plt.xlabel('Sample')
plt.ylabel('Amplitude')
plt.tight_layout()
plt.show()
```



4. Exponential signal

experiment_1 > 4.py > ...

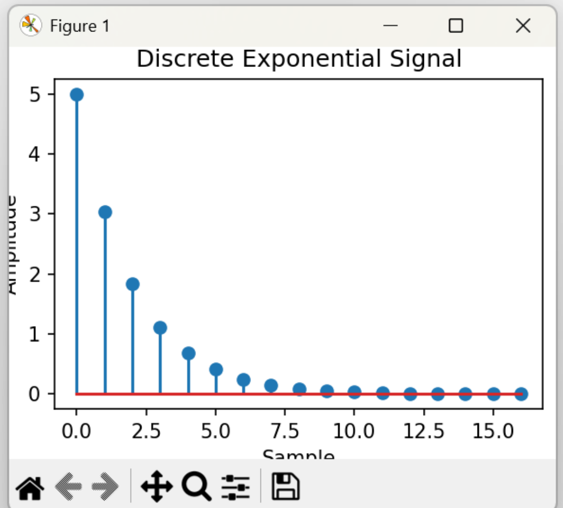
```python
import numpy as np
import matplotlib.pyplot as plt

def simulate_discrete_exponential(num_samples, amplitude, coefficient):
    exponential_signal = amplitude * np.exp(coefficient * np.arange(num_samples))
    return exponential_signal

num_samples = 17
amplitude = 5
coefficient = -0.5

discrete_exponential = simulate_discrete_exponential(num_samples, amplitude, coefficient)

plt.stem(discrete_exponential)
plt.title('Discrete Exponential Signal')
plt.xlabel('Sample')
plt.ylabel('Amplitude')
plt.tight_layout()
plt.show()
```



5. Discrete sine wave

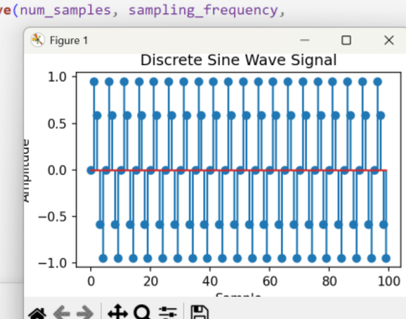
experiment_1 > 5.py > ...

```python
import numpy as np
import matplotlib.pyplot as plt

def simulate_discrete_sine_wave(num_samples, sampling_frequency, amplitude, frequency,
phase):
    time = np.arange(num_samples) / sampling_frequency
    sine_wave = amplitude * np.sin(2 * np.pi * frequency * time + phase)
    return sine_wave

num_samples = 100
sampling_frequency = 10
amplitude = 1
frequency = 2
phase = 0

discrete_sine_wave = simulate_discrete_sine_wave(num_samples, sampling_frequency,
amplitude, frequency, phase)
plt.stem(discrete_sine_wave)
plt.title('Discrete Sine Wave Signal')
plt.xlabel('Sample')
plt.ylabel('Amplitude')
plt.tight_layout()
plt.show()
```



PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
E:\SEM 5\DSIP>python -u "e:\SEM 5\DSIP\experiment_1\5.py"
```

Conclusion : In these experiments, we are using 2 libraries : numpy and matplotlib. After importing, we are defining/creating functions for specific signals : ramp, exponential,ect. After defining function, we are defining/initializing parameters. Then at last we are plotting this using matplot librarty imported earlier by giving appropriate titles on x and y axis.